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DAPR-Net: Symmetric Dual-Stream Pyramid Architecture for Non-Rigid Brain MRI Registration

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ABSTRACT

Deformable image registration is crucial in medical image analysis and is crucial for clinical diagnosis and treatment planning. Traditional methods rely on iterative optimization, which is computationally intensive and slow. Deep learning-based methods, while achieving improved efficiency, are still limited by the local receptive field of convolutions, resulting in poor performance when handling large, non-rigid deformations. Recent approaches combining attention mechanisms with feature pyramids have improved registration accuracy but remain limited in modeling both global and local information and effectively fusing multi-scale features. To overcome these issues, we propose DAPR-Net, an unsupervised registration network based on a two-stream pyramid architecture. Its two-stream encoder combines a local window self-attention mechanism with parallel convolutional branches to simultaneously capture fine details and global contextual information. Multi-level fusion modules are then used to generate multi-scale features. In the decoding stage, the feature pyramid is combined with a multi-scale fusion module to progressively refine the deformation field from coarse to fine levels. Additionally, a multi-scale channel attention mechanism is introduced to better capture fine anatomical details at full resolution. The proposed DAPR-Net is comprehensively evaluated on several publicly available brain MRI datasets. Experimental results demonstrate that our method consistently surpasses existing approaches in terms of key performance metrics, including Dice Similarity Coefficient (DSC), Structural Similarity Index (SSIM), and Peak Signal-to-Noise Ratio (PSNR), highlighting its strong potential for clinical applications.

1. Introduction

Medical Image Registration (MIR) is a key technique in medical image analysis and plays a crucial role in clinical practice, including surgical navigation, computer-aided diagnosis, and radiotherapy[1,2]. Its core objective is to align medical images acquired at different time points, from different imaging modalities, or from different individuals into a unified coordinate system through spatial transformation, thereby achieving spatial consistency of anatomical structures. Although related algorithms have continuously developed in recent years, achieving high-precision registration remains a challenge, especially when large deformations or high-dimensional data are involved.

In deformable image registration, a major challenge is the complex and highly nonlinear spatial deformation relationships between image pairs. These deformations are typically caused by anatomical differences, patient posture changes during image acquisition, or differences in imaging devices. Traditional registration methods[3–6] generally rely on constrained deformation models and achieve accurate registration by repeatedly performing optimization in the diffeomorphic space to maximize image similarity. However, due to the high computational cost and long runtime of the iterative process, these methods have clear limitations when handling complex three-dimensional volumetric data deformations.

With the development of deep learning, convolutional neural network (CNN)-based registration methods have made significant progress in end-to-end modeling. These methods are capable of

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performing effective image registration, but due to the local receptive field limitation of convolution operations, they often fail to adequately capture long-range spatial dependencies and handle large displacement deformations, which can lead to unstable global structural alignment. To address the challenges brought by large deformations, more and more studies have proposed recursive cascaded frameworks or multi-stage models[7–9], as well as multi-resolution or pyramid structures [10–12], to enhance multi-scale deformation representation.

Although pyramid and cascaded structures can improve alignment stability to some extent in large deformation scenarios, these methods often introduce additional computational complexity and may accumulate interpolation errors. Additionally, existing methods often fail to effectively implement cross-scale feature interaction and global context modeling. To overcome these limitations, attention mechanisms have emerged as an effective tool for enhancing long-range dependency modeling. However, the standard self-attention mechanism incurs high computational cost in high-resolution 3D medical images, and excessive reliance on global information may reduce the alignment accuracy of fine anatomical boundaries and local structures. Therefore, how to effectively balance global dependency modeling, cross-scale feature interaction, and local detail preservation under large deformations remains a critical open problem.

To address the above challenges, this paper proposes an unsupervised deformable medical image registration network based on a dual-stream pyramid architecture, referred to as DAPR-Net. Our goal is to resolve the shortcomings of existing methods in large deformation scenarios, particularly in global structural consistency, local detail alignment, and cross-scale feature fusion. Specifically, the proposed method makes innovative contributions in the following aspects:

1. This paper designs a dual-stream encoder architecture that combines local-window self-attention with parallel convolutional branches to simultaneously capture local detailed information and global contextual information. A Multi-level Feature Fusion Block (FFM Block) is further employed to enable cross-scale feature integration, thereby generating powerful multi-scale feature representations for deformation field estimation.
2. This paper constructs a decoding framework based on the Feature Pyramid Network (FPN) to progressively estimate the registration field. A Multi-scale Fusion Block (MSF Block) is further designed to adaptively fuse features from the fixed image and the warped moving image, enhancing the model's ability to capture complex spatial correspondences and handle large-scale anatomical deformations.
3. This paper introduces a Multi-scale Channel Attention Block (MSCA Block) to enhance the model's ability to perceive fine-grained anatomical structures at the full-resolution level, thereby improving the accuracy of capturing complex anatomical structures.

2. Related Work

2.1. Traditional Registration Methods

Traditional image registration methods usually rely on iterative optimization of constrained deformation models to maximize the similarity between the fixed image and the transformed moving image. For example, Avants et al.[3] proposed a symmetric image normalization method that optimizes the registration process by maximizing cross-correlation in the diffeomorphic mapping space. Lorenzi et al.[4] introduced a log-Demons-based diffeomorphic registration algorithm, which achieves accurate alignment by estimating and optimizing a transformation vector field. Gold et al.[5] developed a non-rigid point-set matching technique, namely robust point matching, which adopts spline functions as the transformation model and demonstrates excellent performance in medical image registration tasks involving small deformations. Leordeanu et al.[6] proposed a graph matching algorithm that models graph-structural relationships between corresponding points in two images by constructing an adjacency matrix. Although these iterative optimization methods achieve good registration performance, their main limitation is the high computational cost, which becomes even more pronounced in applications involving three-dimensional volumetric data.

2.2. CNN-based Registration Methods

Early deep learning-based medical image registration methods mainly employed CNNs as the core architecture. CNN-based registration methods are typically built on U-Net backbones and, compared with traditional registration methods, offer end-to-end learning capability and more efficient inference. VoxelMorph[13] was the first to combine U-Net[14] with a spatial transformer network, establishing the first end-to-end trainable deformable image registration framework. However, due to the limitation of the local receptive field of convolution, this architecture is difficult to model global correspondences in large-deformation scenarios, thereby restricting registration performance. To address this issue, some researchers have focused on stacking multiple networks to improve registration accuracy. Kim et al.[7] proposed CycleMorph, which captures transformation relationships at different levels by cascading two registration subnetworks and cyclically switching the order of the fixed and moving images; Pham et al.[8] proposed a recursive cascaded registration network, which progressively deforms images by recursively cascading multiple VoxelMorph models, thereby improving registration accuracy; Hussain et al.[9] further proposed a novel recursive deformable image registration network that integrates hierarchical feature encoding and an adaptive cascaded refinement strategy to achieve accurate registration across multiple resolution scales. Although this method improves registration accuracy under complex deformations, it also leads to the accumulation of interpolation artifacts, which degrades the quality of the deformation field.

To address this problem, some researchers have attempted to optimize deformation field estimation through multi-resolution strategies. Siegert et al.[10] proposed a Laplacian pyramid-based network that effectively enlarges the receptive field and enhances multi-scale deformation modeling capability by progressively fusing coarse-level feature maps. Zhang et al.[11] introduced a hierarchical accumulation mechanism into a Laplacian pyramid network with hierarchical recursion, achieving registration through progressive accumulation of deformation information. Feenstra et al.[12] proposed a local image distance metric based on shape operators, which captures local geometric features via higher-order shape information to enable accurate local structural registration. In addition, Wang et al.[15] proposed the recursive pyramid network RDP, which adopts a progressive recursive strategy to predict deformation fields from coarse to fine; Zhang et al.[16] proposed a hybrid pyramid architecture that effectively improves the accuracy and efficiency of medical image registration through feature-level fusion and iterative registration strategies. Although these methods improve registration performance to some extent, they still have limitations in handling complex deformations and capturing global contextual information.

2.3. Transformer-based Registration Methods

In recent years, Transformers have been widely applied in medical image registration, particularly showing advantages in handling complex deformations and long-range dependencies. The TransMorph method proposed by Chen et al.[17] divides the input image into N patches in the encoding stage, and in the decoding stage, computes long-range relationships among patches through Transformer blocks, thereby aggregating these patches and capturing long-range dependencies between image pairs to improve registration accuracy. Meng et al.[18] proposed a Transformer-based affine registration method, which computes long-range features between image pairs via Transformer blocks

and adopts a multi-scale strategy in the decoding stage to progressively refine the deformation field, further improving registration accuracy.

However, standard self-attention mechanisms still have certain limitations in capturing spatial correspondences between image pairs. To address this issue, Zhang et al.[19] proposed a cross-attention mechanism network that interactively extracts corresponding features from the fixed and moving images; Ghahremani et al.[20] proposed a Vision Transformer registration framework that jointly models short-range and long-range deformation patterns to estimate the deformation field by combining self-attention and cross-scale cross-attention. On this basis, FDR-TransUNet[21] and AutoFuse[22] adopt dual-branch architectures and compute cross-attention between two images through self-attention mechanisms; ModeTv2[23] explicitly models local spatial correspondences using a multi-head neighborhood attention mechanism. In addition, some studies enhance multi-scale representation capability through hierarchical self-attention structures. For example, C2FSwinT[24] and NAP[25] construct coarse-to-fine feature modeling pipelines based on hierarchical Transformers to meet the requirements of multi-scale deformation modeling. Although these attention-based methods provide new opportunities for image registration tasks, their high computational burden and model complexity lead to low efficiency when processing large-scale data. In particular, for large-range deformations and complex anatomical structures, relying solely on global information may ignore local fine-grained features and thus cause instability in registration results.

Despite the considerable progress achieved by existing registration methods, several challenges remain, including high computational cost, limited ability to capture long-range spatial dependencies, insufficient cross-scale feature interaction, and insufficient preservation of fine anatomical details under large deformations. To address these challenges, we propose a dual-stream recursive pyramid network for unsupervised 3D medical image registration. The proposed framework estimates deformation fields in a coarse-to-fine recursive manner, enabling effective global alignment while reducing interpolation error accumulation in cascaded registration. Furthermore, the integration of multi-scale feature fusion and channel attention mechanisms allows the model can more effectively exploit structural information at different scales, enabling more accurate capture of fine-grained anatomical structures and further improving registration accuracy.

3. Methods

3.1. *DAPR-Net Overview*

Overall, DAPR-Net adopts a symmetric dual-stream architecture, consisting of two main components: a dual-stream encoder (Fig. 1 (a)) and a pyramid-structured decoder (Fig. 1 (b)). The dual-stream encoder primarily includes parallel LWSA, convolutional branches, and FFM Block, where the convolutional branch focuses on extracting local detailed information, the LWSA branch models global contextual information, and the FFM Block integrates features from different branches and layers to achieve cross-scale feature fusion, thereby generating more expressive multi-scale feature representations. This design enables the network to learn complementary features from both convolutional and LWSA-based extractors while fully leveraging cross-scale information, thereby enhancing its capability to model complex anatomical structures.

The decoder is composed of multiple submodules, including feature fusion, deformation estimation, and deformation fusion modules. By hierarchically aggregating multi-level features, the decoder performs progressive registration learning at each pyramid level, which facilitates the accurate modeling of large-scale non-rigid deformations. In addition, each decoding level incorporates an MSCA Block and an MSF Block. These components enhance the model's sensitivity to fine-grained local structural variations, effectively mitigating the degradation in registration accuracy that may arise from substantial anatomical discrepancies or large deformation fields. Finally, high-precision alignment of the moving image to the fixed image space is achieved through optimization of the overall loss function, which jointly constrains deformation smoothness and image similarity.

3.2. *Dual-stream Encoder Structure*

The encoder serves as the core feature extraction component of DAPR-Net, responsible for extracting key features from the input images. To enable multi-scale collaborative feature representation, we have designed a symmetric dual-encoder architecture that processes the moving and fixed images in parallel. In each branch of the encoder, the moving and fixed images are separately input into parallel LWSA Blocks and convolutional blocks. These blocks are responsible for extracting global contextual representations and local spatial details, respectively. The LWSA Blocks focus on capturing global information, such as large-scale structures in the image, while the convolutional blocks specialize in extracting fine-grained features, such as edges and small objects. This

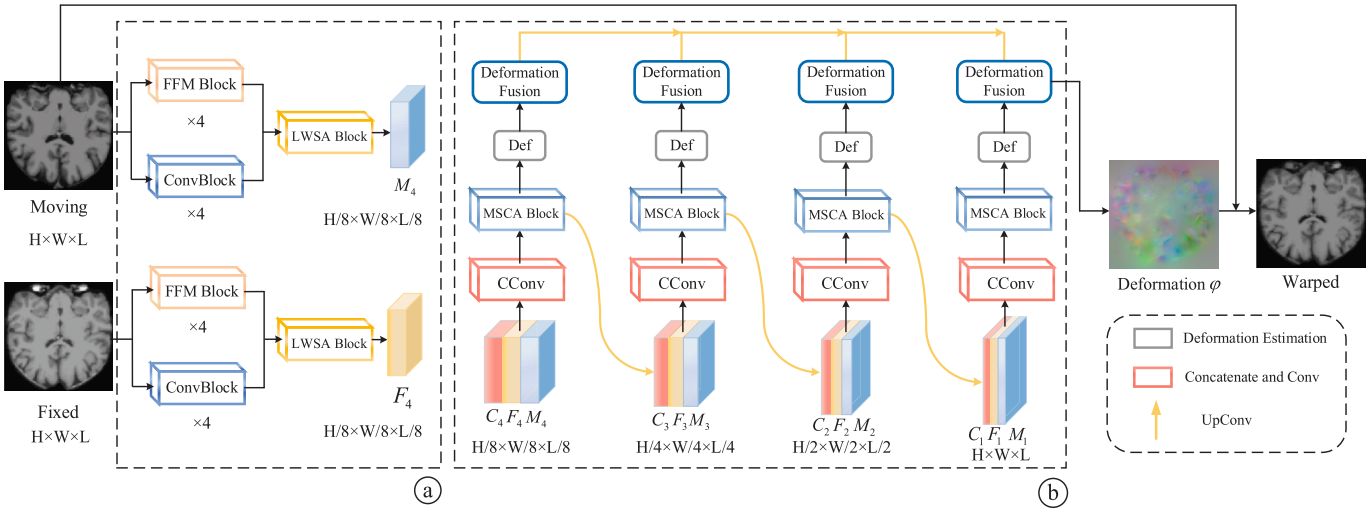


Fig. 1. The network architecture of DAPR-Net.

parallel processing approach helps the model better understand both the overall structure and local features of the images.

Except for the first encoder layer, which preserves the original spatial resolution without downsampling, the subsequent three layers progressively downsample the feature maps. By gradually reducing the resolution of the feature maps, the receptive field is enlarged, allowing the model to capture deformation patterns at multiple scales, thus enhancing its ability to handle large displacements and complex deformations. At the final encoder layer, feature interaction and integration across branches are achieved through the FFM Block. The FFM Block adaptively enhances the correspondence between paired features through channel-wise reweighting and spatial attention mechanisms, while a shared-weight strategy is employed to maintain spatial consistency across modalities. Compared to traditional single-encoder architectures, this dual-encoder design effectively balances global deformation awareness and local detail preservation, demonstrating significant advantages in capturing large displacements and modeling complex non-rigid deformations.

3.2.1. LWSA Block

The LWSA Block[26] is a key component of the dual-stream encoder proposed in this paper and is designed to enhance the collaborative modeling of local details and global structures during feature extraction. In medical image registration tasks, especially under large-deformation scenarios, relying solely on the local receptive field of convolution operations is insufficient to fully model long-range dependencies across regions, thereby affecting the stability of global structural alignment; however, directly adopting standard global self-attention incurs high computational cost on high-resolution 3D medical images. To address this issue, the LWSA Block introduces the window self-attention mechanism of Swin Transformer and extends it to 3D medical image registration tasks. Through local-window and shifted-window strategies, the module can effectively model local details and global structural information while maintaining computational efficiency, and its specific structure is shown in Fig. 2.

The LWSA Block first partitions the input moving image and fixed image into non-overlapping 3D image patches, each with a size of 4 × 4 × 4. Within each image patch, voxel intensity values are flattened into a one-dimensional vector and then mapped to an embedding space of arbitrary dimension through linear projection, thereby forming a token sequence. These tokens are then fed into the LWSA Block for local window partitioning and self-attention computation.

The LWSA Block adopts the conventional window partitioning scheme, uniformly dividing feature blocks into non-overlapping local windows, where each window has a size of $P_x \times P_y \times P_z$. Within each local window, a self-attention mechanism is used to extract features among individual image patches. To capture relationships between patches in adjacent windows, the shifted-window design in Swin Transformer is employed. Specifically, each window is shifted along the $(P_x/2, P_y/2, P_z/2)$ direction by a predefined number of patches, thereby forming new local windows that partially overlap with neighboring regions. Based on these operations, the computation process of the LWSA Block can be described as follows:

$$\hat{z}_i = W\text{-}MSA(LN(z_{i-1})) + z_{i-1} \tag{1}$$

$$z_i = MLP(LN(\hat{z}_i)) + \hat{z}_i \tag{2}$$

$$\hat{z}_{i+1} = SW\text{-}MSA(LN(\hat{z}_i)) + z_i \tag{3}$$

$$\hat{z}_{i+1} = W\text{-}MSA(LN(\hat{z}_{i+1})) + \hat{z}_{i+1} \tag{4}$$

Where $W\text{-}MSA(\cdot)$ and $SW\text{-}MSA(\cdot)$ denote the window-based and shifted-window-based multi-head self-attention modules, respectively; $MLP(\cdot)$ denotes the multilayer perceptron module; and $LN(\cdot)$ denotes layer normalization. In the LWSA Block, z_{i-1}, z_i and z_{i+1} correspond to the input, the intermediate features after W-MSA, and the output, respectively.

By introducing the local-window and shifted-window strategies, the LWSA module can efficiently capture both local and global features. Its design addresses the difficulty of traditional methods in capturing long-range dependencies under large-deformation scenarios, thereby improving registration accuracy and stability.

3.2.2. FFM Block

The FFM Block[25] is located at the last layer of the DAPR-Net encoder and is designed to enhance the interaction between local perceptual features and global semantic features, thereby further extracting more discriminative fused features. In the dual-stream

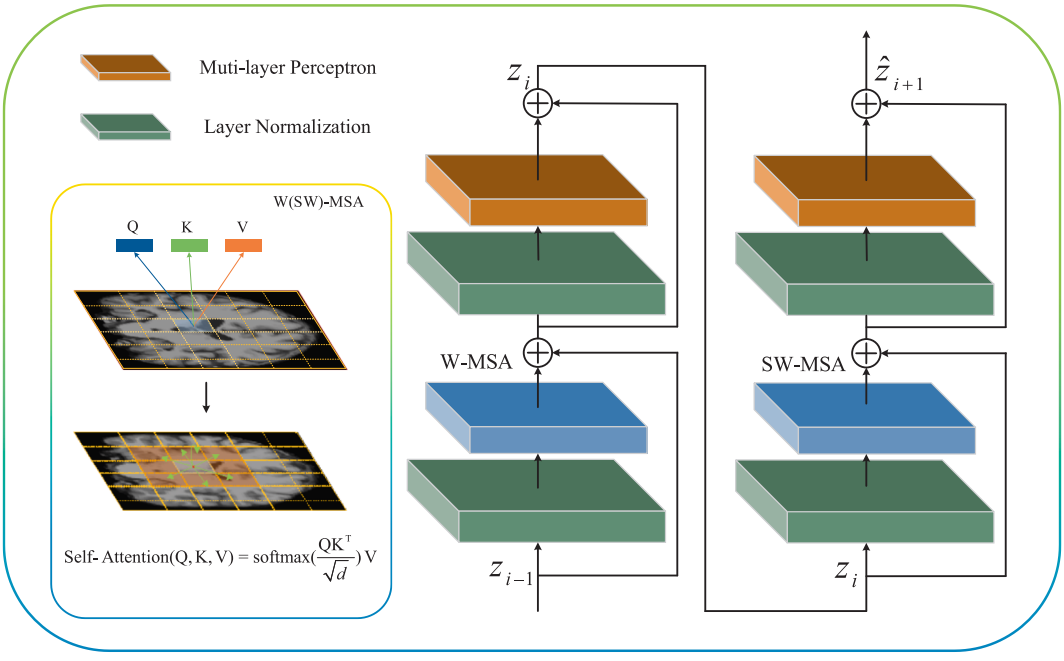


Fig. 2. LWSA Block Architecture Diagram.

encoder, the CNN branch is good at extracting fine-grained local textures and boundary information, whereas the LWSA branch is more suitable for modeling long-range dependencies and global context. However, these two types of features differ in representation form and response distribution, and direct concatenation or summation may easily introduce redundant information and weaken their complementary advantages. Therefore, this paper introduces the FFM Block to achieve effective alignment and fusion of cross-branch features, and its structure is shown in Fig. 3.

The FFM Block first uses a $1 \times 1 \times 1$ convolution to adjust the dimensions of intermediate features from the CNN network, aligning them with the intermediate features T_i from the LWSA Block. Although CNN and LWSA possess complementary characteristics, direct concatenation or summation might introduce redundant information. Therefore, the FFM first applies Channel Attention (CA) to each of the two feature maps separately to adaptively adjust the weights of each channel. Subsequently, it introduces a Cross-domain Fusion (CFB) block, which utilizes a multi-head self-attention mechanism to achieve cross-fusion between features, thereby obtaining cross-domain fused features.

Specifically, for feature maps C_i' and T_i' that have been processed by CA, depth-wise separable convolutions are used to map them into a set of query vectors Q, key vectors K, and value vectors V, respectively. Then, two cross-attention mechanism calculations are performed, and the specific formulas are as follows:

$$\mathrm{Attn}_x = \mathrm{softmax}\left(\frac{Q_y K_x^T}{\sqrt{d_k}}\right) V_x \tag{5}$$

$$\mathrm{Attn}_y = \mathrm{softmax}\left(\frac{Q_x K_y^T}{\sqrt{d_k}}\right) V_y \tag{6}$$

Where d_k denotes the scaling factor, and denotes the dimension of each attention head. By computing their similarity, cross-domain fusion between features is achieved. Next, matrix multiplication is used to obtain features enhanced with cross-domain correlations, further strengthening key information between feature maps while suppressing irrelevant features. Finally, these different features are concatenated to form preliminary fused features. The fused features are then fed into the feature fusion module to further extract effective information and reduce redundant outputs.

Through cross-domain fusion and channel attention mechanisms, the FFM Block can effectively integrate local details and global semantic information. Through the collaborative effect of the feature fusion module, this design enables DAPR-Net to effectively combine the fine-grained local features extracted by the CNN branch with the global contextual information captured by the LWSA Block, thereby forming more powerful and more discriminative feature representations and enhancing the network's ability to perform accurate image registration.

3.3. Recursive Pyramid Decoder

To improve the accuracy of deformation field prediction and enhance image registration performance, the decoder recursively fuses multi-level features and, by combining a pyramid structure with a multi-scale attention mechanism, gradually optimizes the deformation estimation.

Specifically, the architecture consists of four decoding stages. At each decoding level, the deformation prediction at the current resolution is progressively refined through multiple iterative steps. In the l-th layer, the fixed and moving images are first fed into the MSCA Block (as shown in Fig. 4 (a)) to generate a fused feature map C_l. Subsequently, the fused feature is input into the Deformation Estimation Module (Def). This module predicts the deformation field corresponding to the current resolution level, enabling accurate alignment of the moving image with the fixed image.

To further enhance the learning capability for non-rigid deformations, while simultaneously improving the overall registration performance and spatial consistency, we designed the MSCA Block, whose specific structure is shown in Fig. 4 (b).

By fusing deformation fields from different levels, features at different scales can be captured more effectively. Especially when dealing with large-scale non-rigid deformations, coarse deformation fields can help fine deformation fields converge faster. Therefore, in each step, we fuse the newly estimated deformation field φ_s with the previously obtained deformation field φ_{s-1},and after upsampling, input it into the next decoder layer to generate a more refined deformation field.

3.3.1. MSCA Block

The MSCA Block[27] is a lightweight and efficient attention mechanism originally proposed for spatial information encoding in semantic segmentation tasks. During the recursive pyramid decoding process, the deformation field needs to be refined layer by layer, while the sensitivity of different spatial locations to the registration result varies significantly. If feature propagation relies only on conventional convolutions, the model tends to assign similar response weights to different regions, making it difficult to effectively highlight key anatomical boundaries and regions with significant deformation. Therefore, in this paper, we adaptively optimize the structure of MSCA for medical image

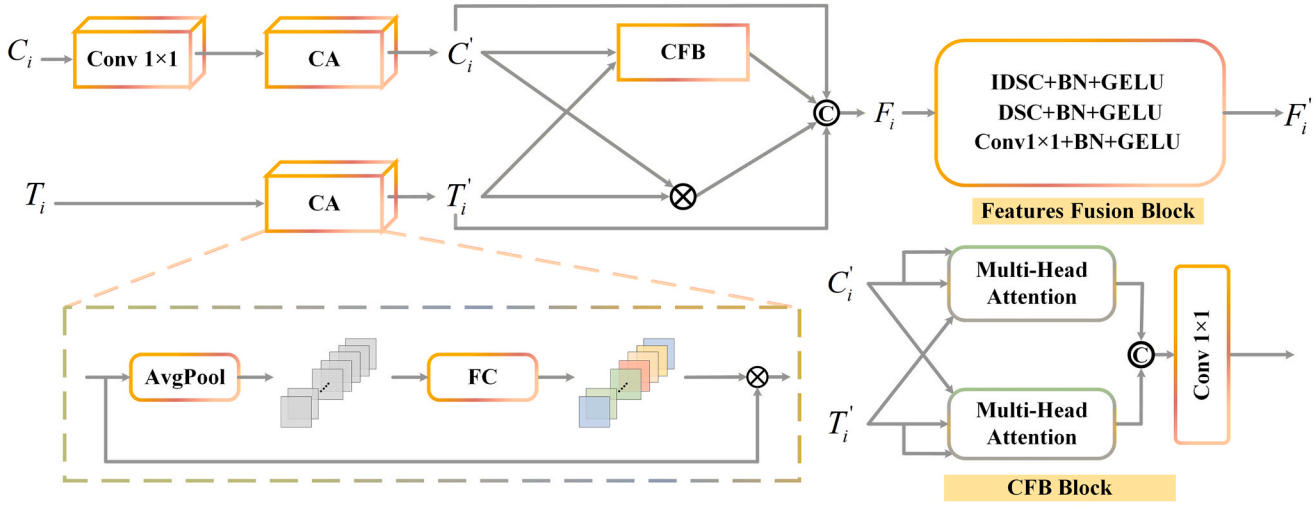


Fig. 3. FFM Block Architecture Diagram.

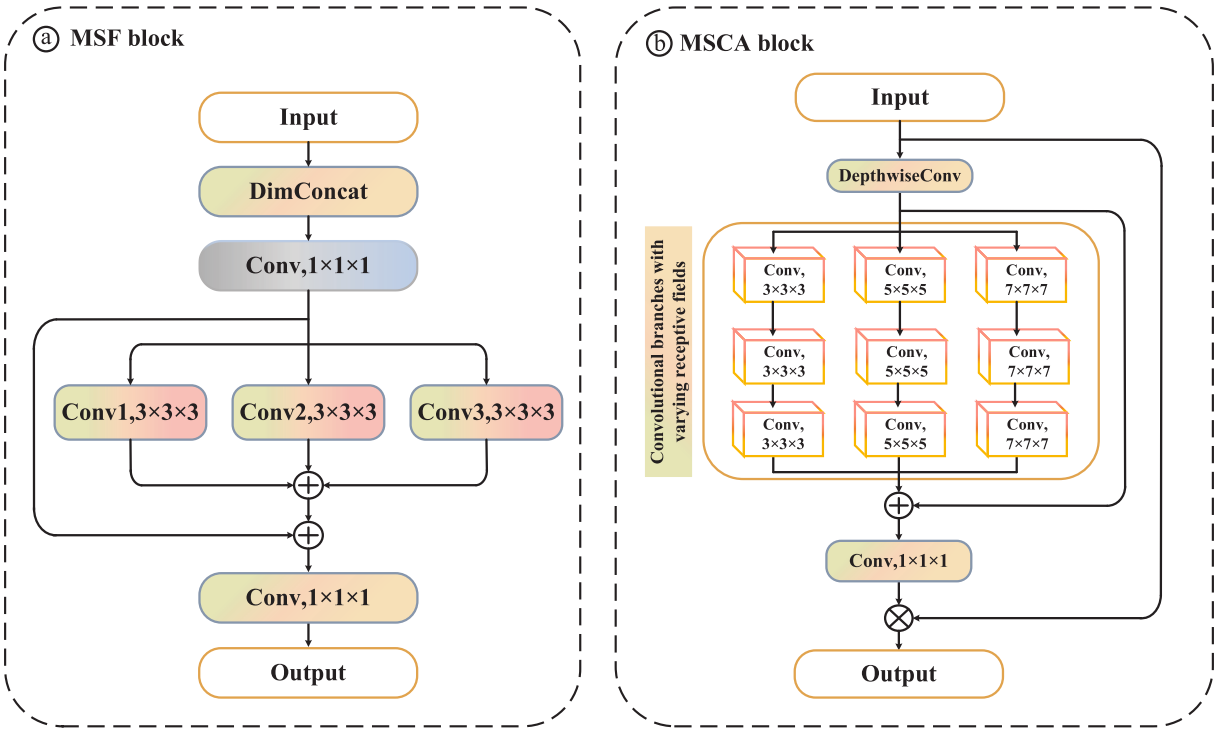


Fig. 4. MSF Block and MSCA Block Architecture Diagram.

registration tasks and integrate it into the recursive pyramid decoder to enhance the model's ability to model non-rigid deformations, and its specific structure is shown in Fig. 4 (b).

MSCA Block first uses Depth-wise Convolution to process the input feature to extract local contextual information. This process helps preserve the spatial structure of the original image and lays the foundation for subsequent multi-scale feature extraction. Subsequently, it approximates large-kernel depthwise convolutions through multiple convolutional branches with different receptive fields $3 \times 3 \times 3$, $5 \times 5 \times 5$, and $7 \times 7 \times 7$. These multi-scale convolutional branches are capable of concurrently capturing both fine-grained local details and broader global contextual information. Following the extraction of multi-scale features, the MSCA Block performs a weighted fusion of these features. Subsequently, an attention map is generated via a $1 \times 1 \times 1$ convolutional layer, enabling the model to emphasize salient regions. This attention map reflects the importance weights of different spatial locations and is directly applied to the original input feature, achieving feature re-weighting via element-wise multiplication:

$$\mathrm{M} = Conv_{1\times1}\left(\sum_{i=0}^{3} Scale_i(DW - Conv(F))\right) \tag{7}$$

$$\mathrm{M}' = Att \otimes \mathrm{F} \tag{8}$$

Where F represents the input feature map. $\mathrm{DW}-Conv(\cdot)$ denotes the depth-wise convolution operation, which is used to extract local contextual information and preserve the original spatial structure. $\mathrm{Scale}_i(\cdot)$ represents the learnable scaling factor of the i-th scale branch, which is used to fuse feature responses of different receptive fields. M is the attention map, reflecting the importance weight of each spatial position. M' is the reweighted output feature map.

In the medical image registration framework proposed in this paper, MSCA is embedded into the decoding process of each level. By combining MSCA with a recursive pyramid structure, the deformation field prediction can be gradually refined, and the response intensity of each spatial location can be dynamically adjusted through the attention mechanism, so that the model pays more attention to the deformation-sensitive regions.

3.3.2. MSF Block

In medical image registration tasks, the Spatial Transformer is one of the core components for modeling image deformation. It receives an input image and a predicted flow field, and spatially deforms the image according to the flow field through a sampling mechanism, thereby achieving pixel-level mapping between the reference image and the target image. Standard Spatial Transformers are usually based on affine transformations or non-rigid deformation models and can accomplish basic spatial alignment. However, in complex non-rigid deformation scenarios, relying only on a fixed flow field obtained from a single prediction is often insufficient to fully characterize subtle local structural offsets, which may easily lead to inadequate registration in local regions. To enhance the adaptive modeling capability of the decoding stage for complex spatial correspondences, we design an MSF Block, and its structure is shown in Fig. 4 (a).

The MSF Block first applies a $1 \times 1 \times 1$ convolutional layer to reduce the channel dimension of feature map F. The compressed feature map is then simultaneously fed into three parallel $3 \times 3 \times 3$ convolutional branches, each designed to extract local features at different orientations and spatial locations. The features extracted by each branch maintain consistent channel dimensions and are fused via element-wise summation, forming a feature representation with multi-scale perception capabilities. Finally, the fused feature map passes through another $1 \times 1 \times 1$ convolutional layer to restore to the target output channel count, forming the final output. The MSF Block predicts additional offsets and adds them to the original flow field, thereby enhancing spatial transformations:

$$\varphi_{out} = \varphi + \Delta\varphi = \varphi + \mathrm{MSF}(\mathrm{F}) \tag{9}$$

Where F represents the input feature tensor. $\Delta\varphi$ represents the additional offset predicted by MSF Block, φ_{out} represents the final corrected flow field used for image transformation.

The MSF Block introduces a residual offset compensation mechanism, enabling the model to continuously refine the existing flow field and progressively correct local alignment errors during coarse-to-fine

decoding. This mechanism not only enhances the model's adaptive adjustment capability for complex spatial correspondences, but also helps alleviate the accumulation effect of local mismatches in large-deformation scenarios, thereby further improving registration accuracy and stability under complex non-rigid deformations.

4. Experiment

4.1. Dataset

This study employs two publicly available brain MRI datasets, LPBA [28] and IXI[17], to comprehensively evaluate the proposed algorithm. The LPBA dataset contains 40 T1-weighted MRI images, which have been pre-registered to the MNI305 standard space via rigid transformation and cropped to a size of 160 × 192 × 160 (voxel size 1 mm^3). Among these, 30 volumetric images are used for training (yielding 30 × 29 training image pairs), and 10 volumetric images are used for testing (yielding 10 × 9 test image pairs).The IXI dataset provides both original and FreeSurfer[29] affine-registered data, resampled to 160 × 192 × 224 (voxel size 2 mm^3).Training used 407 vol images (which can generate 407 training image pairs), and testing used 58 vol images (which can generate 58 test image pairs), and validation used 115 vol images (which can generate 115 validation image pairs).

4.2. Evaluation Metrics

To provide a comprehensive assessment of registration performance, this study utilizes three evaluation metrics: Dice Similarity Coefficient (DSC), Structural Similarity Index Measure (SSIM), and Peak Signal-to-Noise Ratio (PSNR). These metrics evaluate the similarity between registered images from multiple perspectives. Specifically, DSC quantifies the spatial overlap between regions of interest in the fixed image and the deformed moving image, with values closer to 1 indicating higher registration accuracy. PSNR reflects the degree of image distortion, with higher values indicating less distortion and better preservation of structural details. SSIM evaluates similarity by jointly considering consistency in brightness, contrast, and structural information, values approaching 1 denote higher perceptual similarity.

In addition, to assess the quality of the predicted non-rigid deformation field, we introduce the metric J_ϕ— the percentage of voxels with non-positive Jacobian determinants (commonly referred to as "folded voxels"). A J_ϕ value closer to 0 implies fewer spatially distorted or topologically invalid regions in the deformation field, indicating higher field quality and smoother, more biologically plausible deformations. Through this multi-faceted evaluation framework, the performance of the registration network can be accurately and comprehensively quantified.

4.3. Implementation Details

All experiments were conducted on an NVIDIA RTX 6000 Ada Generation GPU equipped with 48 GB memory. The software environment consisted of PyTorch 1.10.0, CUDA 11.3.0, and Python 3.8 to ensure efficient training of the proposed model. During training, the Adam optimizer was adopted, and the batch size was set to 1. The learning rate was kept at 0.0001 throughout the entire training process to ensure stable convergence and improve algorithm performance and model generalization capability.

4.4. Comparison Methods

We compared the proposed method with several state-of-the-art registration approaches. Specifically, the comparison methods include: (1)SyN[30]: a classical traditional registration algorithm, evaluated using the SyNOnly configuration in ANTs. (2)VoxelMorph[13]: a widely used single-stage deep learning-based registration network. (3) TransMorph[17]: a single-stage registration network that incorporates a Swin Transformer to enhance the encoder's feature representation capabilities. (4) TransMatch[26]: a symmetric dual-stream registration network that performs explicit multi-level feature matching through multi-level local window cross-attention modules, facilitating more accurate image registration. (5)AutoFuse[22]: a symmetric two-stream registration network that utilizes an adaptive multi-level fusion gating mechanism for registration. (6)XMorpher[31]: a Transformer-based network architecture that adopts cross-attention mechanisms and weighting strategies to fuse multi-level feature maps and deformation fields. (7) ModeT[32]: a pyramid network that utilizes the multi-head attention mechanism of Transformer to explicitly decompose and model multiple motion patterns for fusion to generate deformation fields. (8)Im2Grid[33]: a pyramid deformation framework that predicts deformation fields at each level using neighborhood attention. (9)PAN [34]: a two-stream pyramid network with channel attention, which generates deformation fields through a decoder based on the multi-head local attention mechanism.

4.5. Experimental Results

4.5.1. Comparison of Experimental Results

The results presented in Table 1 demonstrate that DAPR-Net outperforms all compared methods on the IXI dataset across multiple evaluation metrics, including DSC, PSNR, and SSIM. In addition, the proposed method also achieves the best performance in terms of deformation field quality (%$|J_\phi| < 0$). These results indicate that DAPR-Net is able to produce more accurate and reliable registration outcomes compared with the competing approaches. Fig. 5 further illustrates the qualitative registration results of different methods on the IXI dataset, providing a visual comparison of the alignment performance achieved by each model.

To further analyze the performance differences among the models at the sample level, we computed and visualized pixel-level difference maps between images before and after registration, as shown in Fig. 6 and Fig. 7. Specifically, the pre-registration difference map represents the absolute intensity difference between the moving image and the fixed image before the registration transformation is applied. In contrast, the post-registration difference map quantifies the absolute intensity difference between the fixed image and the warped moving image after the registration process. These visualizations provide an intuitive way to assess the effectiveness of the registration results from a

Table 1
Performance comparison of registration accuracy of different methods on the IXI dataset.

Models	Metrics DSC	PSNR	SSIM	%$\|J_\phi\| < 0$
SyN[30]	0.723 ± 0.027	31.998 ± 0.869	0.985 ± 0.003	< 0.0049%
Im2Grid[33]	0.732 ± 0.029	32.213 ± 0.889	0.986 ± 0.004	< 0.0055%
VoxelMorph [13]	0.668 ± 0.065	30.905 ± 0.856	0.982 ± 0.004	< 0.0061%
TransMatch [26]	0.718 ± 0.027	31.679 ± 0.869	0.985 ± 0.003	< 0.0083%
XMorpher[31]	0.720 ± 0.018	32.021 ± 0.996	0.986 ± 0.003	< 0.0068%
AutoFuse[22]	0.742 ± 0.022	32.476 ± 0.968	0.987 ± 0.003	**< 0.0006%**
ModeT[32]	0.743 ± 0.043	36.631 ± 1.320	0.988 ± 0.003	< 0.0062%
TransMorph [17]	0.710 ± 0.055	32.213 ± 0.889	0.986 ± 0.003	< 0.0092%
PAN[34]	0.711 ± 0.032	32.134 ± 0.899	0.986 ± 0.002	< 0.0018%
DAPR-Net (Ours)	**0.748 ± 0.041**	**36.391 ± 1.350**	**0.995 ± 0.002**	< 0.0034%

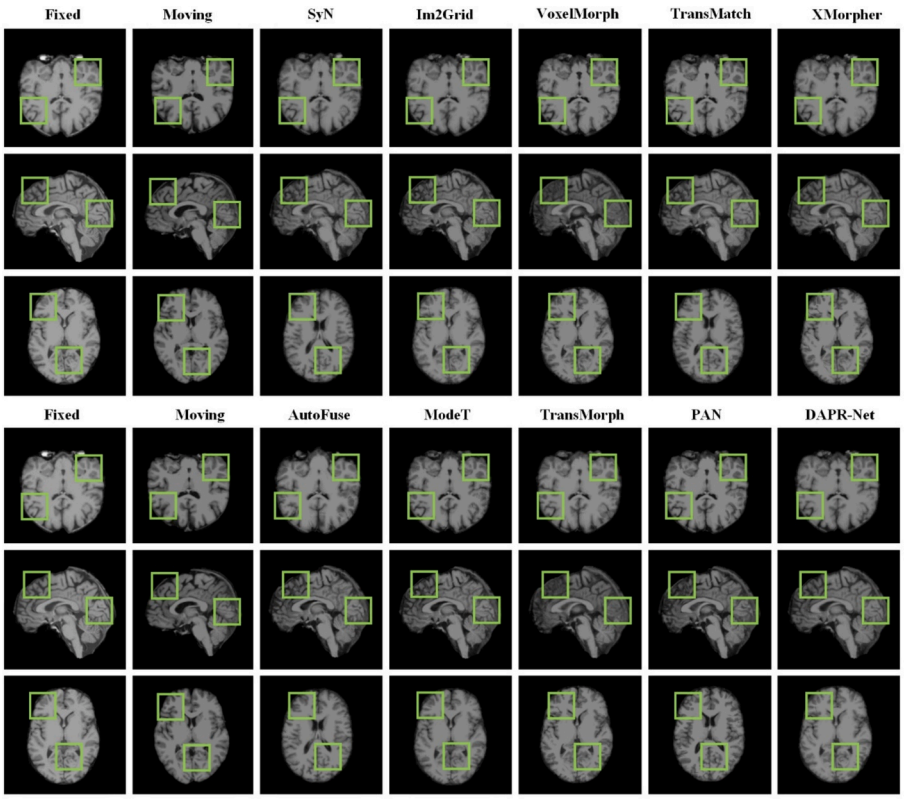


Fig. 5. Visualization of registration results and corresponding deformation fields of different methods on the IXI dataset.

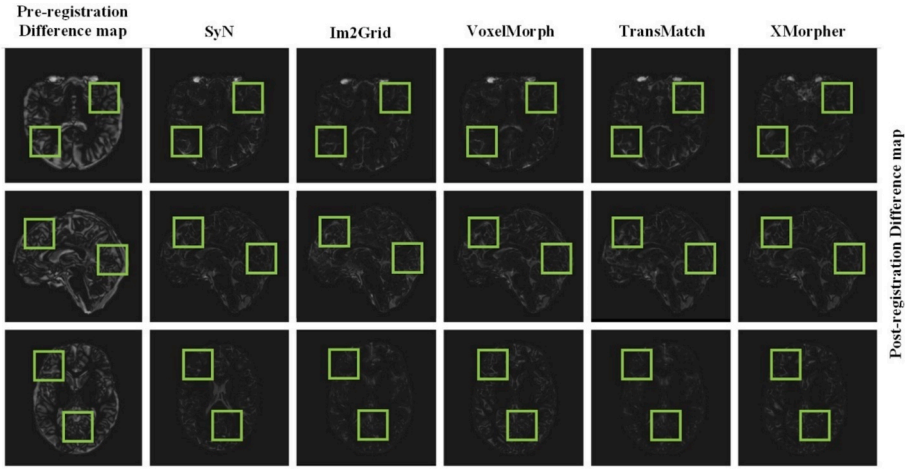


Fig. 6. Difference maps before and after registration of different methods on the IXI dataset.

pixel-wise perspective. In general, smaller difference values indicate more accurate spatial correspondence and better alignment between the two images.

The experimental results demonstrate that the difference maps between the registered images generated by our network and the fixed images exhibit improved visual consistency and overall visual quality. In particular, the difference signals are significantly reduced in the edge regions of complex anatomical structures, where accurate alignment is typically more challenging to achieve. This reduction suggests that the proposed method is able to better capture subtle structural variations and preserve important anatomical details during the registration process. These observations further confirm that DAPR-Net achieves more precise voxel-level alignment, effectively reduces registration errors, and demonstrates superior performance compared with the other evaluated methods.

4.5.2. Ablation Experiment Results

In this study, we conducted ablation experiments on the IXI dataset to investigate the individual contributions of the dual encoder, the MSCA Block, and the MSF Block. The corresponding results are summarized in Table 2. When no additional module was introduced, the DSC was 0.732. After introducing the dual encoder, the DSC increased to 0.737, validating that the dual-stream structure effectively enhances the network's ability to perceive correspondences between image pairs by independently extracting and fusing features from the fixed and moving images. Based on the dual-stream encoder, further adding the MSCA Block increased the DSC to 0.743. This improvement indicates that the MSCA Block can dynamically focus on regions with complex anatomical

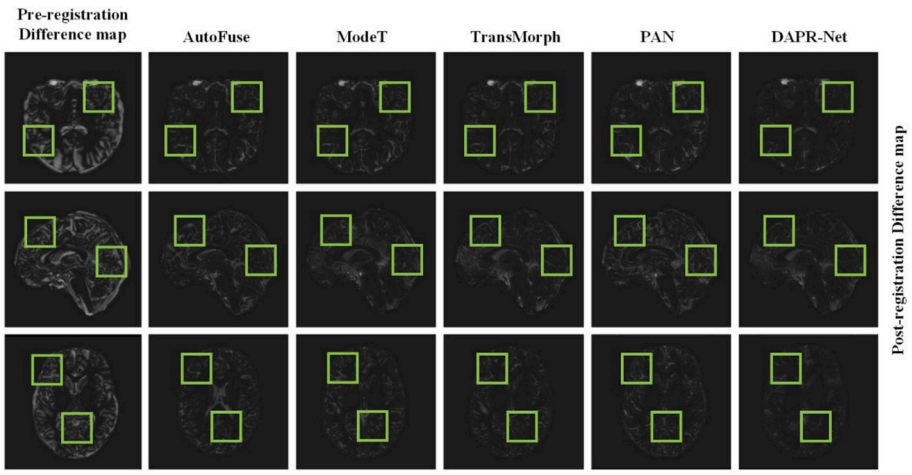


Fig. 7. Difference maps before and after registration of different methods on the IXI dataset.

Table 2
Results of ablation experiments.

Dual Encoder	MSCA Block	MSF Block	DSC	$\%\lvert J_\phi\rvert < 0$
×	×	×	0.732 ± 0.035	< 0.0018%
√	×	×	0.737 ± 0.040	< 0.0012%
√	√	×	0.743 ± 0.042	< 0.0009%
√	√	√	0.748 ± 0.041	< 0.0034%

structures and enhance the perception of local details, thereby improving registration accuracy. Finally, after integrating the MSF Block into the network, the DSC reached the maximum value of 0.748. Meanwhile, the deformation field quality index ($\%\lvert J_\phi\rvert \leq 0$) also remained at a very low level, and the generated deformation field was smooth and reasonable. Overall, the combination of the three modules achieves the best results in the image registration task.

4.5.3. Generalization Experiment Results

To comprehensively evaluate the generalization capability of the proposed DAPR-Net, we not only conducted extensive training on the IXI dataset but also assessed its performance on the LPBA dataset. This cross-dataset validation is intended to examine the adaptability and robustness of the model when confronted with data exhibiting different anatomical characteristics, imaging conditions, and intensity distributions.

As summarized in Table 3, DAPR-Net demonstrates consistently superior quantitative performance among multiple state-of-the-art registration methods. In particular, the model achieves significant improvements on key evaluation metrics such as the DSC and PSNR. These improvements indicate that the proposed framework can more accurately preserve anatomical structural consistency and maintain high image fidelity after registration, thereby reflecting the robustness and stability of the learned deformation representations.

In addition, as shown by the visual results in Fig. 8, DAPR-Net exhibits stronger capability on the LPBA dataset in accurately aligning fine-grained brain anatomical structures. The deformation fields predicted by our network are spatially smooth and topologically consistent, without obvious folding artifacts. Moreover, the difference maps between the registered images and the fixed images show lower residual intensity, further verifying the accuracy and smoothness of the registration results.

In summary, these experimental results strongly demonstrate that the feature extraction mechanism and registration strategy learned by DAPR-Net possess strong generalization capability. The model can effectively transfer its learned representations to previously unseen datasets, thereby maintaining robust and reliable registration performance across different imaging domains.

Table 3
Performance comparison of registration accuracy of different methods on the LPBA dataset.

Models	Metrics DSC	PSNR	SSIM	$\%\lvert J_\phi\rvert < 0$
SyN[30]	0.704 ± 0.023	31.485 ± 0.193	0.995 ± 0.003	< 0.0002%
Im2Grid[33]	0.710 ± 0.034	31.210 ± 0.074	0.995 ± 0.002	< 0.0003%
VoxelMorph [17]	0.682 ± 0.015	32.092 ± 1.442	0.990 ± 0.002	< 0.00097%
TransMatch [26]	0.697 ± 0.014	31.422 ± 1.350	0.996 ± 0.003	< 0.00013%
XMorpher[31]	0.726 ± 0.022	32.108 ± 0.531	0.995 ± 0.002	< 0.0096%
AutoFuse[22]	0.695 ± 0.018	32.094 ± 1.475	0.990 ± 0.003	< 0.00062%
ModeT[32]	0.729 ± 0.017	31.916 ± 0.307	0.995 ± 0.002	< 0.00032%
TransMorph [17]	0.689 ± 0.017	31.673 ± 0.428	0.995 ± 0.003	**< 0.0001%**
PAN[34]	0.711 ± 0.019	31.153 ± 1.352	0.995 ± 0.002	< 0.00012%
DAPR-Net (Ours)	**0.731 ± 0.017**	**32.246 ± 1.580**	**0.996 ± 0.002**	< 0.00015%

4.5.4. Challenging Case Analysis

Although DAPR-Net achieves strong overall performance across multiple datasets and evaluation metrics, it is important to analyze cases where the model performs less favorably to more comprehensively understand its behavior and limitations. As shown in Table 1, while DAPR-Net achieves the best DSC, PSNR, and SSIM on the IXI dataset, the deformation field quality metric based on the Jacobian determinant is not optimal. Specifically, the percentage of non-positive Jacobian determinants is slightly higher than that of some competing methods, such as AutoFuse, indicating that locally compressed regions may occur in certain complex deformation scenarios.

To further investigate this, we conducted a detailed analysis of representative challenging samples with large deformations. Fig. 9 visualizes the deformation grids generated by different methods on these difficult cases. In regions with moderate anatomical variations, all methods, including DAPR-Net, generate smooth and regular deformation grids. However, in areas with large anatomical displacement or complex structural boundaries, such as cortical folding regions or areas near ventricular boundaries, the deformation grids produced by DAPR-Net are locally denser, reflecting local compression in the deformation

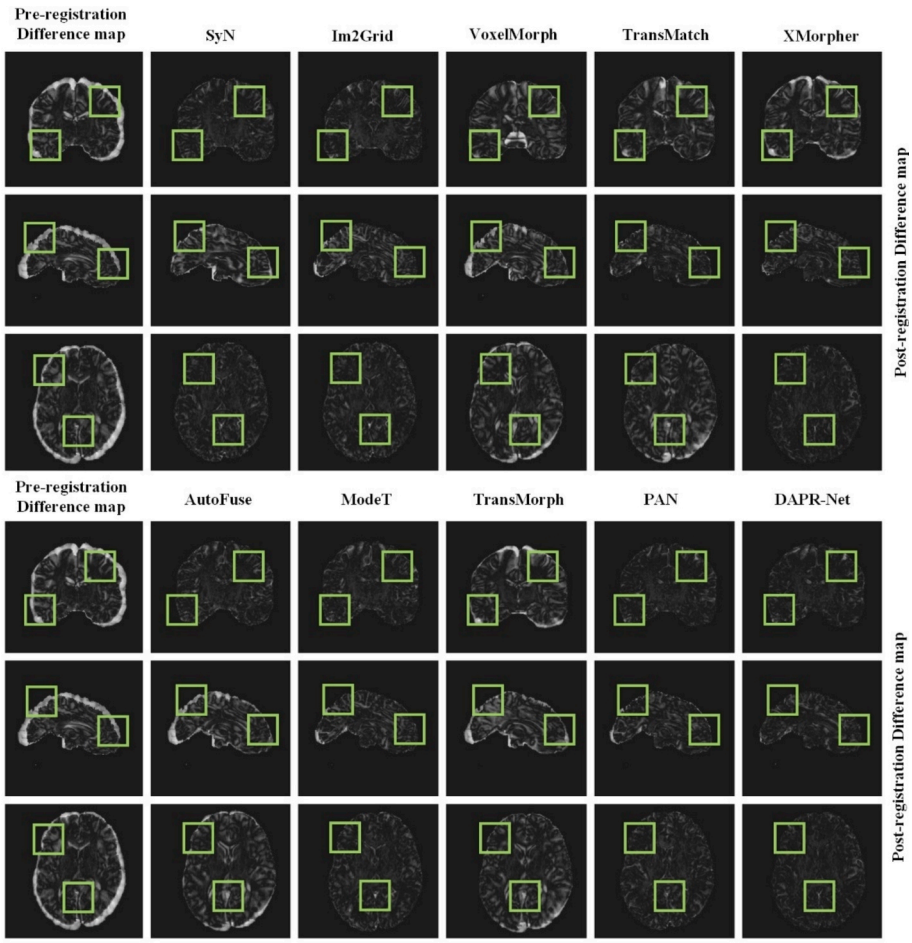


Fig. 8. Difference maps before and after registration of different methods on the LPBA dataset.

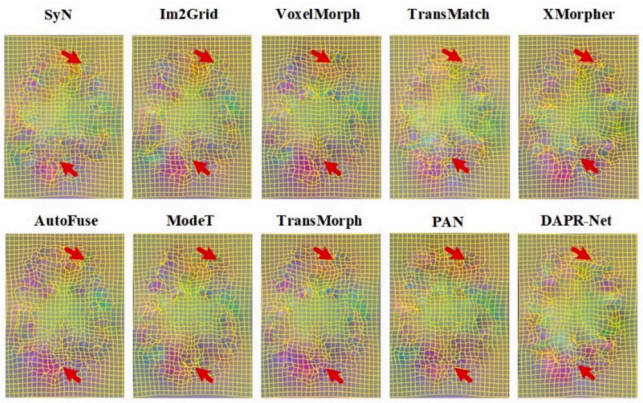


Fig. 9. Comparison of deformation field grids generated by different methods.

field.

This behavior can be explained by the design of DAPR-Net. The dual-stream encoder integrates LWSA and convolutional features, which enables accurate modeling of both global and local dependencies. The FFM and MSCA Blocks further enhance cross-scale feature integration and local detail preservation. While these components improve overall alignment and registration stability, the network prioritizes accurate global and local structural consistency, which may occasionally result in slight local compression in regions with extreme deformation. This explains why the Jacobian-based metric is slightly lower in some challenging cases.

Despite this, the deformation fields generated by DAPR-Net maintain good topological consistency without obvious folding or grid crossing, and the difference maps and registration results demonstrate that anatomical structures between the fixed and warped images remain well aligned. Therefore, DAPR-Net provides stable and accurate registration in most scenarios, while the observed local compression highlights areas for potential future improvement, such as introducing stronger topology-preserving constraints.

5. Discussion

The dual-stream pyramid network DAPR-Net proposed in this study demonstrates superior registration performance on the IXI and LPBA datasets, achieving significant improvements in key metrics such as DSC, PSNR, and SSIM over commonly used deep learning models and classical methods. DAPR-Net employs dual-stream encoders to independently extract image features and integrates an FFM Block to achieve cross-branch fusion and multi-scale feature generation, effectively mitigating feature confusion in single-stream structures. Compared with existing methods, it performs better in modeling both global and local information and handling large deformations. The MSCA Block captures multi-scale contextual information and dynamically focuses on complex anatomical regions (e.g., sulci and gyri) using a channel attention mechanism, while the MSF Block fuses multi-scale features to enhance the modeling of complex nonlinear deformations. Ablation studies show that each core module complements the others, collectively improving registration accuracy, and these advantages are visually evident in Figs. 6-8, enabling DAPR-Net to effectively handle complex 3D brain MRI images.

The practical significance of this method is not only reflected in improved registration accuracy and enhanced capability for handling large deformations, but also in its broad potential for real-world applications. In medical imaging, accurate image registration provides clinicians with more precise anatomical structural information, thereby improving the safety and effectiveness of surgical navigation. Especially in brain surgery and radiotherapy, DAPR-Net can effectively align anatomical structures and provide more accurate data support for treatment planning and surgical strategy formulation. In addition, due to its strong capability, DAPR-Net also provides possibilities for further promoting precision in the field of medical imaging, particularly in complex cases requiring highly accurate alignment.

Although DAPR-Net demonstrates state-of-the-art performance, its relatively large number of model parameters may impose certain limitations on memory usage and computational speed. In some extreme cases, such as those involving complex brain boundaries or large-scale deformation regions, there is still room for further optimization of the registration performance.

6. Conclusion

This paper proposes DAPR-Net, a dual-stream pyramid network for unsupervised brain MRI registration. Through a carefully designed dual-stream encoder, MSCA Block, and MSF Block, the network achieves collaborative perception of global structure and local features, and dynamically fuses multi-scale information within a coarse-to-fine recursive pyramid framework to generate high-quality deformation fields. Extensive experiments on public datasets, including IXI and LPBA, demonstrate that MADReg performs excellently in terms of registration accuracy and deformation field plausibility, with strong generalization capability. Future work could focus on reducing model parameters through compression techniques or optimizing the network architecture to improve computational efficiency, while also enhancing the ability to capture fine-grained local details, thereby further improving registration accuracy and expanding application scenarios.

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Data availability

The datasets used in the article are all available. The IXI dataset is available at https://brain-development.org/ixi-dataset/. The LPBA dataset is available at Atlases Downloads – LONI Resource.

CRediT authorship contribution statement

Xin Yang: Writing – review & editing, Funding acquisition. **Runhan Hou:** Writing – review & editing, Writing – original draft. **Liwei Deng:** Writing – review & editing, Funding acquisition. **Jing Wang:** Writing – review & editing. **Sijuan Huang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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